

Intermediate Macroeconomics

BSc. in Business Administration and BSc. in Finance and Accounting

Intermediate Macroeconomics – BMAK

Chapter 2: Macroeconomic Measurement

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A Motivating Example: The meaning of GDP

In Chapter I, we had seen that comparisons across multiple decades or longer of levels of “income per capita (per person)” across regions or countries motivate the study of (long-run) economic growth. We had also seen that comparisons across a few years of “growth rates of output” within regions or countries motivate the study of business cycles.

The numbers reported in these tables and figures in Chapter I were based on the so-called **National Income and Product Accounts (NIPA)**. As we will learn in the first part of the current chapter, the NIPA provide the most widely used framework for measuring the state of macroeconomic activity, and comparing it across time periods and/or countries/regions. Understanding the key elements of the NIPA is also important so as to appreciate what precisely is meant by terms such as “income” or “output”, and what interpretations these terms may be given.

A first key measure of the state of the economy is **gross domestic product (GDP)**. GDP may be determined in three different ways:

(i) Product Approach

GDP

- = value of all goods and services newly produced, minus the value of all intermediate goods and services used in the production process
- = sum of value added at each stage of the production process.

When measuring economic activity through the level of production, we need to ensure that the value of the same product is not counted more than once. For example, a manufacturer of a laptop may use a processor produced by another manufacturer, and this processor should not be counted as both a good produced on its own, and as a component of the laptop. By excluding at each stage of production the cost of intermediate goods and services, the value-added approach ensures that there is no “double counting”.

Data Examples

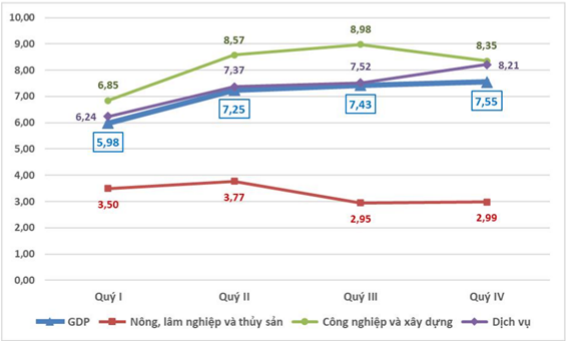
BÁO CÁO TÌNH HÌNH KINH TẾ – XÃ HỘI QUÝ IV VÀ NĂM 2024

1. Tốc độ tăng tổng sản phẩm trong nước

Tổng sản phẩm trong nước (GDP) quý IV/2024 ước tính tăng 7,55% so với cùng kỳ năm trước, chỉ thấp hơn quý IV các năm 2017, 2018 trong giai đoạn 2011-2024[1], duy trì xu hướng quý sau cao hơn quý trước (quý I tăng 5,98%, quý II tăng 7,25%, quý III tăng 7,43%). Trong đó, khu vực nông, lâm nghiệp và thủy sản tăng 2,99%, đóng góp 4,86% vào mức tăng tổng giá trị tăng thêm toàn nền kinh tế; khu vực công nghiệp và xây dựng tăng 8,35%, đóng góp 44,03%; khu vực dịch vụ tăng 8,21%, đóng góp 51,11%. Về sử dụng GDP quý IV/2024, tiêu dùng cuối cùng tăng 7,54% so với cùng kỳ năm trước; tích lũy tài sản tăng 7,98%; xuất khẩu hàng hóa và dịch vụ tăng 11,35%; nhập khẩu hàng hóa và dịch vụ tăng 13,49%.

GDP năm 2024 ước tính tăng 7,09% so với năm trước, chỉ thấp hơn tốc độ tăng của các năm 2018, 2019 và 2022 trong giai đoạn 2011-2024[2]. Trong mức tăng tổng giá trị tăng thêm toàn nền kinh tế, khu vực nông, lâm nghiệp và thủy sản tăng 3,27%, đóng góp 5,37%; khu vực công nghiệp và xây dựng tăng 8,24%, đóng góp 45,17%; khu vực dịch vụ tăng 7,38%, đóng góp 49,46%.

Hình 1. Tốc độ tăng GDP và VA các khu vực theo quý năm 2024 (%)



Data examples

Economic Sector	Germany in 2023		USA [USD billion]	
	Billions of EUR	%	Billions of USD	%
Agriculture, Forestry, and Fishing	38.7	0.9	222.3	0.8
Industry (Excluding Construction)	1252.6	29.1	5017.5	18.9
Construction	172.2	4.0	1060	4.0
Trade, Transport, Accommodation, and Food Service	688.8	16.0	2650	10.0
Information and Communication	193.4	4.5	1590	6.0
Financial and Insurance Services	172.2	4.0	2120	8.0
Real Estate Activities	430.5	10.0	3180	12.0
Business Services	516.6	12.0	3445	13.0
Public Services, Education, and Health	645.8	15.0	2385	9.0
Other Services	193.4	4.5	2120	8.0
Net Taxes (Taxes Less Subsidies) on Products	301.4	7.0	2650	10.0
Gross Domestic Product (Total)	4305	100.0	26500	100.0

Data Examples: Vietnam in 2024

1. Tổng sản phẩm trong nước theo giá hiện hành	Tỷ đồng					
	Số bộ quý II năm 2024	Ước tính quý IV năm 2024	Cộng dồn cả năm 2024	Cơ cấu (%)		Cả năm 2024
				Quý II năm 2024	Quý IV năm 2024	
TỔNG SỐ	288498	329740	11511867	100.00	100.00	100.00
Nông, lâm nghiệp và thủy sản	339433	415226	1365709	11.72	12.59	11.86
Nông nghiệp	244922	323190	1023329	8.40	9.80	8.89
Lâm nghiệp	14592	17409	56814	0.50	0.53	0.51
Thủy sản	79919	74827	285566	2.78	2.26	2.46
Công nghiệp và xây dựng	1119474	1245248	4333235	38.67	37.76	37.64
Công nghiệp	995425	1022837	3640105	32.36	31.02	31.82
Khai khoáng	65306	69038	287334	2.26	2.09	2.50
Công nghiệp chế biến, chế tạo	726277	920791	2812396	25.09	24.89	24.43
Sản xuất và phân phối điện, khí đốt, nước nóng, hơi nước và điều hòa không khí	129982	117294	483555	4.46	3.56	4.20
Cung cấp nước; hoạt động quản lý và xử lý rác thải, nước thải	14960	15714	60806	0.52	0.48	0.49
Xây dựng	183050	222411	690132	6.32	6.74	6.02
Dịch vụ	1203831	1386287	4876381	41.56	42.04	42.36
Bán buôn và bán lẻ; sửa chữa ô tô, mô tô, xe máy và xe có động cơ khác	270011	300892	1122833	9.36	9.12	9.75
Vận tải, kho bãi	140854	154807	586199	4.87	4.69	5.17
Dịch vụ lưu trú và ăn uống	78040	78013	291095	2.83	2.37	2.53
Thông tin và truyền thông	96834	108789	387489	3.34	3.30	3.37
Hoạt động tài chính, ngân hàng và bảo hiểm	143537	173821	559387	4.96	5.27	4.83
Hoạt động kinh doanh bất động sản	90389	123089	395976	3.33	3.73	3.44
Hoạt động chuyên môn, khoa học và công nghệ	81618	86500	237421	2.83	2.02	2.06
Hoạt động hành chính và dịch vụ hỗ trợ	34014	33107	138291	1.17	1.00	1.21
Hoạt động của Đảng, Cộng sản, tổ chức chính trị-xã hội; quản lý Nhà nước, an ninh quốc phòng; đảm bảo xã hội bất bược	54836	78228	233019	1.89	2.37	2.02
Giáo dục và đào tạo	109177	135136	450609	3.77	4.10	3.91
Y tế và hoạt động trợ giúp xã hội	82141	89308	304708	2.84	2.71	2.85
Nghệ thuật, vui chơi và giải trí	18575	20941	71488	0.57	0.63	0.62
Hoạt động dịch vụ khác	17707	20899	76676	0.61	0.63	0.67
Hoạt động làm thuê các công việc trong các hộ gia đình, sản xuất sản phẩm vật chất và dịch vụ tự tiêu dùng của hộ gia đình	3460	3708	14019	0.12	0.11	0.12
Thuế sản phẩm trừ thuế cấp sản phẩm	23228	23647	93652	8.03	7.80	8.14

The source of 'money' to pay workers and firm owners (all of them have to pay taxes) is the value added during the production process.

GDP = sum of all incomes earned in the production of all new goods and services that constitute value added.

GDP by Income Categories: Germany

(Billion Euro and Percent of GDP, 2023)

Labor Income (Compensation of Employees)	€2,062.3 billion	47.9%
Capital Income (Gross Operating Surplus)	€1,584.0 billion	36.8%
Self-Employed Income (Mixed Income)	€176.0 billion	4.1%
Taxes and Subsidies (Net Taxes on Production and Imports)	€482.7 billion	11.2%
Gross Domestic Product	€4,305 billion	100.0%

Source: Eurostat (2024)

GDP = sum of all expenditures by final users of newly produced goods and services.

$$Y = C + I + G + EX - IM \quad (1)$$

- **C (Consumption):** Total spending by private households on newly produced goods and services (excluding new housing).
- **I (Investment):** Total spending by private firms on newly produced capital goods, including equipment, structures, and inventories.
- **G (Government Spending):** Total spending by the public sector on goods and services (excluding transfer payments).
- **EX (Exports):** Value of newly produced goods and services sold to foreign markets.
- **IM (Imports):** Value of foreign-produced goods and services purchased domestically.

This equation is referred to as the Gross National Expenditure (GNE) equation.

GDP by Expenditure Components: Germany

(Billion Euro and Percent of GDP, 2023)

Component	Billion Euro	Percent of GDP
Final Consumption Expenditure	1900.5	53.0%
Gross Fixed Capital Formation	780.2	21.8%
Machinery and Equipment	260.0	–
Construction	390.5	–
Other Products	140.0	–
Changes in Inventories	-8.5	–
Government Final Consumption Expenditure	730.0	20.4%
Trade Balance	210.0	5.8%
Exports	1700.0	–
Imports	1490.0	–
Gross Domestic Product	3580.7	100.0%

91 Sử dụng tổng sản phẩm trong nước theo giá hiện hành

Gross domestic product by expenditure category at current prices

	2019	2020	2021	2022	Sơ bộ Prel. 2023
Tỷ đồng - Bill. dong					
TỔNG SỐ - TOTAL	7707200	8044386	8487476	9621372	10320311
Tích lũy tài sản - <i>Gross capital formation</i>	2464760	2567421	2792006	3111600	3266027
Tài sản cố định - <i>Gross fixed capital formation</i>	2340104	2435664	2634076	2940853	3107212
Thay đổi tồn kho - <i>Changes in inventories</i>	124656	131757	157930	170747	158815
Tiêu dùng cuối cùng ⁽¹⁾ - <i>Final consumption⁽¹⁾</i>	5118113	5264720	5625696	6109397	6518276
Nhà nước - <i>State</i>	738260	762512	813532	848501	913096
Hộ dân cư - <i>Household</i>	4379853	4502208	4712164	5260896	5605180
Chênh lệch xuất, nhập khẩu hàng hoá và dịch vụ <i>Trade balance (goods & services)</i>	432448	443836	86961	354705	837676
Sai số - <i>Statistical discrepancy</i>	-308121	-231591	82813	45670	-301668

As we have seen in the data, when calculating GDP using the product, income, or expenditure approach, we always obtain the same total value for GDP. This reflects that for the aggregate economy, **production equals income equals expenditure.**

Let us further illustrate this fact and learn about some key rules of NIPA accounting through considering examples of GDP calculations:

Calculating the Effects of a Single Market Transaction on GDP

On January 20, **2025**, a domestic household purchases from a grocery store two liters of milk, at **1.5 Euro per liter**. The grocery store purchased the milk five days earlier at a wholesale price of **1 Euro per liter**.

How to calculate the effect of this January 20 transaction on the year **2025** GDP of the domestic economy under the product, income, and expenditure approaches?

The January 20 transaction contributes to the year **2025** GDP of the domestic economy as follows:

- **Product approach:** Sale of milk on January 20 creates value added produced in the amount of:
 - ▶ **3 Euro** (value of milk as sold by the grocery store)
 - ▶ **– 2 Euro** (value of milk as purchased by the grocery store: intermediate good)
 - ▶ **= 1 Euro.**
- **Income approach:** Sale of milk on January 20 creates value added earned in the amount of:
 - ▶ **(3 Euro – 2 Euro) = 1 Euro**
 - ▶ (Income recipients may include staff and owners of the grocery store, as well as the government, if there was a sales tax involved.)
- **Expenditure approach:**
 - ▶ Purchase of milk on January 20 is consumption expenditure in the amount of **3 Euro.**
 - ▶ **At the same time,** the value of the grocery store's inventory through this purchase is reduced by **2 Euro** (negative inventory investment).
 - ▶ Making use of Equation (1),

$$Y = C + I + G + TB,$$

Calculating GDP for a Sample Economy

Quiet Muffler GmbH has a factory in Landshut valued at **2,000,000 Euro**. During the year **2025**, Quiet Muffler builds **ten mufflers**. The market value of each muffler is **800 Euro**. Quiet Muffler's costs for the production of each muffler are:

- Labor: **300 Euro**
- Interest on debt: **250 Euro**
- Imported components from Malaysia: **100 Euro**
- Taxes: **50 Euro**

Ecological Engine GmbH, located in Ingolstadt, has a factory valued at **4,000,000 Euro**. During **2025**, Ecological Engine builds **eleven engines**, each with a market value of **5,000 Euro**. The costs to produce each engine are:

- Labor: **2,000 Euro**
- Imported components from Romania: **1,000 Euro**
- Taxes: **1,000 Euro**

Ecological Engine sells **ten** of these engines, but at the end of the year **2025**, it had not sold the eleventh engine, which was put into inventory.

Quiet Muffler sells all **ten mufflers**, and Ecological Engine sells all **ten engines** to **Future Vehicles AG**, located in Stuttgart. Using Quiet Muffler's mufflers and Ecological Engine's engines, Future Vehicles builds **ten cars** in **2025** at a cost of **15,000 Euro** each:

- Components: **5,800 Euro**
- Labor: **6,200 Euro**
- Taxes: **3,000 Euro per car**

Future Vehicles sells **seven** of these cars to German households for **22,000 Euro each**, while the remaining **three cars** are put into inventory.

How to calculate (under the product, income, and expenditure approaches) the year **2025** GDP of this sample German economy consisting of Quiet Muffler, Ecological Engine, and Future Vehicles?

• GDP Based on the Product Approach:

We need to determine the value added by each firm in **2025**, and then sum the values added across all firms in the sample economy (the following amounts are in Euro):

	Quiet Muffler	Ecological Engine	Future Vehicle	Total
Value of Goods and Services Produced	10×800	$10 \times 5,000$ $+1 \times 4,000$	$7 \times 22,000$ $+3 \times 15,000$	261,000
Less Intermediate Goods & Services	-10×100	$-11 \times 1,000$	$-10 \times 5,800$	-70,000
Total = Value Added GDP (Product)	7,000	43,000	141,000	191,000

- **GDP Based on the Income Approach:**

We need to add up all (value-added) incomes earned in the sample economy in **2025**. Note that one type of income earned is the firms' profits. We calculate these in the first step, as **profit = revenue – cost**:

Profit Quiet Muffler

<i>Revenue</i>	10×800	$= 8,000$
<i>Cost</i>		
Labor	10×300	$= 3,000$
Parts (imported)	10×100	$= 1,000$
Interest payments	10×250	$= 2,500$
Tax payments	10×50	$= 500$
<i>Profit</i>		$= 1,000$

Other types of value-added income earned in the sample economy in **2025** include labor income (of the employees), interest income (of the banks), and tax revenue (of the government).

Adding up all value-added incomes earned in the sample economy in **2025**:

	Quiet Muffler	Ecological Engine	Future Vehicles	Total
Profits	1,000	10,000	49,000	60,000
Labor Income	3,000	22,000	62,000	87,000
Interest Income	2,500	0	0	2,500
Tax Revenue	500	11,000	30,000	41,500
Total = GDP Based on Income Approach				191,000

In addition to GDP, measures of the state of the economy often analyzed include **gross national product (GNP)** and **gross national disposable product (GNDP)**.

GNP is related to GDP as follows:

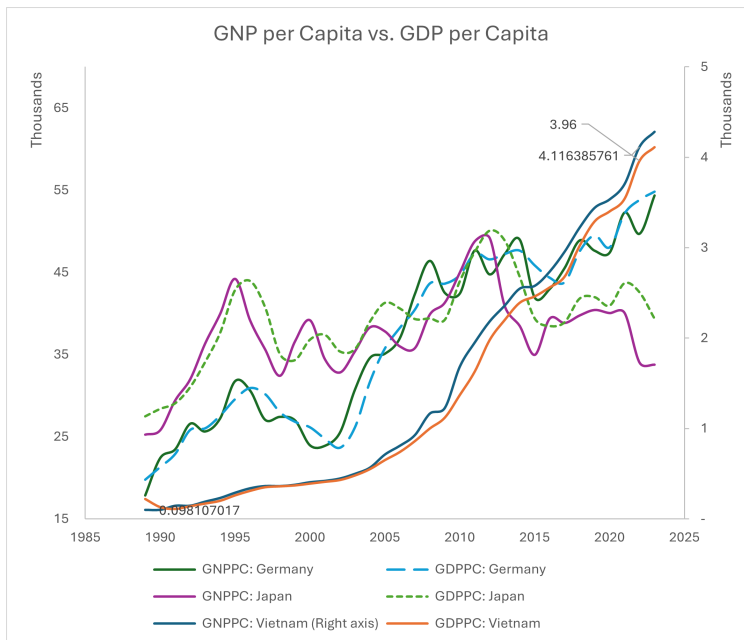
$$\left\{ \begin{array}{l} \text{GDP} \\ + \text{ primary income received by residents from nonresidents} \\ - \text{ primary income paid by residents to nonresidents} \\ = \text{GNP} \end{array} \right\} \text{ balance on primary income (BPI)}$$

Notes:

- Primary income received from nonresidents represents the return received by residents for providing to nonresidents temporary use of labor, financial resources, or nonproduced nonfinancial assets (the latter include, for example, licenses and leases).
- Examples include compensation of employees, interest income, dividends, and rents, all as received by residents from nonresidents.

Thus, GDP captures all incomes received by the producers of new goods and services operating within the borders of the domestic country (irrespective of where their permanent residence is), whereas GNP captures all incomes received by the producers of new goods and services who are permanent residents of the domestic country (irrespective of where they received these incomes).

On terminology: GNP is often referred to as **GNI (gross national income)**. BPI is sometimes referred to as **net factor income from abroad (NFIA)**.



GNP and GNDP Relationships:

$$\begin{array}{lcl}
 \text{GDP} & & \\
 + \text{ primary income received by residents from nonresidents} & \} & \text{balance on primary income (BPI)} \\
 - \text{ primary income paid by residents to nonresidents} & & \\
 = \text{GNP} & &
 \end{array}$$

$$\begin{array}{lcl}
 \text{GNP} & & \\
 + \text{ secondary income received by residents from nonresidents} & \} & \text{balance on secondary income (BSI)} \\
 - \text{ secondary income paid by residents to nonresidents} & & \\
 = \text{GNDP} & &
 \end{array}$$

Notes:

- Primary income received from nonresidents represents the return received by residents for providing to nonresidents temporary use of labor, financial resources, or nonproduced nonfinancial assets (e.g., licenses, leases).
- Examples include compensation of employees, interest income, dividends, and rents, all as received by residents from nonresidents.
- Secondary income received from nonresidents represents income received by residents without providing to nonresidents a corresponding economic value in return.
- Examples include personal transfers, social contributions, insurance premiums, and insurance claims, all as received by residents from nonresidents.

Terminology: GNP is often referred to as **GNI (gross national income)**. GNDP is often referred to as **GNDI (gross national disposable income)**. BPI is sometimes referred to as **net factor income from abroad (NFIA)**. BSI is sometimes referred to as **net unilateral transfers (NUT)**.

Distinction Between Gross and Net Measures

Subtracting technical and physical depreciation of the stock of physical capital, we move from gross measures to the corresponding net measures. For example:

$$\begin{array}{lcl}
 \text{GDP} & & \\
 - \text{ consumption of fixed capital (depreciation)} & & \\
 = \text{NDP (net domestic product)} & &
 \end{array}$$

Net measures convey how much income is available if the value of the productive capacities (that is, the stock of physical capital) is to be maintained.

Calculating a Firm's Contributions to GDP, GNP, GNDP, and NDP

In the year 2025, Berlin-based InVogue develops a new app listing current sales events in several German cities; the app in the year 2025 is downloaded globally 500,000 times at a purchase price of 3 Euro. InVogue's IT equipment at the beginning of the year 2025 is valued at 120,000 Euro.

InVogue pays the following annual wages to its programmers: 550,000 Euro to programmers who are residents of Germany, and 150,000 Euro to programmers who are residents of Poland. The annual rate of depreciation on InVogue's **IT** equipment is 25%.

The programmers who are residents of Germany in 2025 transfer 5,000 Euro of their earnings to an orphanage in Cambodia. The programmers who are residents of Poland in 2025 transfer 20,000 Euro of their earnings to relatives in Poland. The profits of InVogue are paid as dividends to its owners, who are residents of Qatar.

Calculate the year 2025 contributions of InVogue to the GDP, GNP, GNDP, and NDP of the German economy.

The Year 2025 Contributions of InVogue to the German Economy

- GDP Contribution: Value added in the amount of 3 Euro (sales price of app) $\times$ 500,000 (number of downloads) = 1,500,000 Euro.

- GNP Contribution: Both the earnings of the Polish programmers (150,000 Euro) and the dividend payments earned by the Qatari owners (which in turn are equal to the profits of InVogue:

$$1,500,000 - 700,000 = 800,000 \text{ Euro}$$

constitute primary income paid to nonresidents. The GNP contribution is thus:

$$1,500,000 \text{ Euro (GDP contribution)} - 950,000 \text{ Euro (BPI contribution)} = 550,000 \text{ Euro}$$

- GNDP Contribution: The transfer of the German programmers to the Cambodian orphanage (5,000 Euro) constitutes secondary income paid to nonresidents. The GNDP contribution is thus:

$$550,000 \text{ Euro (GNP contribution)} - 5,000 \text{ Euro (BSI contribution)} = 545,000 \text{ Euro}$$

- NDP Contribution: The 2025 depreciation is equal to:

$$120,000 \times 0.25 = 30,000 \text{ Euro}$$

Thus, the NDP contribution is:

$$1,500,000 \text{ Euro (GDP contribution)} - 30,000 \text{ Euro (depreciation)} = 1,470,000 \text{ Euro}$$

● **Measurement issues:**

- ▶ Some goods and services, for example home production, are not transacted and are thus not captured in GDP.
- ▶ Some goods and services are not officially transacted, for example informal or illegal activities, and are thus difficult to capture as part of GDP.
- ▶ Changes in the quality of goods and services are often ignored, especially for services, and may be difficult to capture.

● **Conceptual issues:**

- ▶ Some components of GDP may be viewed as "regrettables" (not enhancing the well-being of society).
- ▶ GDP does not take into account factors other than income that are important for the well-being of society, for example:
 - ★ Value of health,
 - ★ Value of leisure time, and
 - ★ Value of the environment.

- Alternative measures of a society's standard of living exist, for example the "Human Development Index" (HDI), calculated by the United Nations.
- HDI takes into account information on health (such as life expectancy at birth) and education (such as years of schooling), in addition to GDP, in aiming to measure the "quality of life".
- Despite the existence of such alternative measures, GDP continues to be the main measure used in practice for comparing the standard of living achieved across time periods and countries.
- Countries with higher income still enjoy better education, nutrition, and health care.
- Countries with higher income still enjoy better environment, have larger conversations and has "enough buffer zone" to respect the environment (Say, they don't eat wild animal often).
 - ... Higher income allows improvement in different aspects of life.
 - ... People who criticize GDP measurement often come from the rich countries, where they already enjoy very high standard of living.
 - ... Yet, look at what is happening in Europe: You only value your things when they are gone!
- **We do not grow at any costs, of course!**

Current account - Cán cân vãng lai

Understanding the interactions of the domestic economy with other economies and the rest of the world requires knowledge of the so-called Balance of Payments (BoP) accounts.

The BoP is composed of three sub-accounts:

- Current account
- Capital account
- Financial account

Current Account – Cán cân vãng lai captures the following transactions between residents and nonresidents:

- Exports and imports of goods and services
- Primary income received and paid
- Secondary income received and paid

Current account balance (CA) is given by:

$$CA = TB + BPI + BSI \quad (2)$$

Capital Account – Cán cân vốn

The capital account captures the following transactions between residents of the domestic economy and nonresidents:

- Acquisition and disposal of nonproduced nonfinancial assets.
(Including, for example, land, water, licenses, leases, brand names, trademarks and domain names.)
- Capital transfers received and granted.
(Capital transfers, in contrast to transfers that are part of secondary income, involve the transfer of ownership of nonproduced nonfinancial assets or the forgiveness of liabilities by a creditor without receiving a corresponding value in return; this includes, for example, the donation of land and debt forgiveness.)

Capital account balance (KA) is given by:

$$KA = -NANA + NRCT \quad (3)$$

Financial Account – Cán cân tài chính

The financial account captures financial asset and financial liability transactions between residents of the domestic economy and nonresidents, categorized as follows:

- Foreign direct investment (Including, for example, equity, bonds and non-bank loans that entitle to enterprise voting power of at least 10%)
- Portfolio investment (Including, for example, equity, bonds and non-bank loans that entitle to enterprise voting power of less than 10%, as well as derivatives)
- Other investment (Including, for example, currency, bank deposits and bank loans)
- Reserves (Including, for example, foreign currency reserves held by the monetary authorities and foreign currency loans granted by international organizations to the monetary authorities)

Financial Account Balance (FA)

The **financial account balance (FA)** is given by:

$$FA = NAFA + NIFL \quad (4)$$

If $FA > 0$, the domestic economy in net terms supplies funds to the rest of the world, and is called a **net lender**:

Its sum of acquisition of financial assets and repayment of financial liabilities exceeds its sum of disposal of financial assets and incurrence of financial liabilities.

If $FA < 0$, the domestic economy in net terms borrows funds from the rest of the world, and is called a **net borrower**:

Its sum of disposal of financial assets and incurrence of financial liabilities exceeds its sum of acquisition of financial assets and repayment of financial liabilities.

Recording Principles I

- In recording transactions in the current, capital and financial accounts, the BoP invokes the principle of **doubly-entry accounting**: Each transaction is recorded in the BoP of the domestic economy as consisting of two entries of equal value, **credit** and **debit**, reflecting the inflow and outflow elements of each transaction:
- Any transaction leading to a payment from nonresidents to residents is first recorded as a credit entry, and the second recording is then as an offsetting debit entry.
- Any transaction leading to a payment from residents to nonresidents is first recorded as a debit entry, and the second recording is then as an offsetting credit entry.
- **Thus**, the BoP transactions for which the first entry is a **credit entry** are: exports of goods and services, primary and secondary income payments received, disposal of nonproduced nonfinancial assets, capital transfers received, disposal of financial assets, as well as incurrences of financial liabilities.

Recording Principles II

- Also, the BoP transactions for which the first entry is a **debit entry** are: imports of goods and services, primary and secondary income payments made, acquisition of nonproduced nonfinancial assets, capital transfers granted, acquisition of financial assets, as well as repayment of financial liabilities.

Recording Principles III

- **Per se** separate is the issue of the sign of each entry. The sign of the first and second entries is in line with the account balances given in Equations (2), (3) and (4):
- Thus, transactions that enter with a **plus sign** are: exports of goods and services, primary and secondary income payments received, disposal of nonproduced nonfinancial assets, capital transfers received, acquisition of financial assets, as well as repayment of financial liabilities.
- Also, transactions that enter with a **minus sign** are: imports of goods and services, primary and secondary income payments made, acquisition of nonproduced nonfinancial assets, capital transfers granted, disposal of financial assets, as well as incurrences of financial liabilities.
- It follows that in the **current and capital accounts**, the first entries that are credit entries are recorded with a **plus sign**, and the first entries that are debit entries are recorded with a **minus sign**.
- However, in the **financial account**, the first entries that are credit entries are recorded with a **minus sign**, and the first entries that are debit entries are recorded with a **plus sign**.

Let us summarize these BoP accounting principles:

	Credits	Debits	Balance
Current Account			
	Exports of goods and services (+) Primary income received (+) Secondary income received (+)	Imports of goods and services (-) Primary income paid (-) Secondary income paid (-)	TB BPI BSI CA
Capital Account			
	Disposal of nonproduced nonfinancial assets (+) Capital transfers received (+)	Acquisition of nonproduced nonfinancial assets (-) Capital transfers granted (-)	– NANA NRCT KA
Financial Account			
	Disposal of foreign direct investment assets (-) Incurrences of foreign direct investment liabilities (-)	Acquisition of foreign direct investment assets (+) Repayment of foreign direct investment liabilities (+)	NAFA (FDI) – NIFL (FDI)
	Disposal of portfolio investment assets (-) Incurrences of portfolio investment liabilities (-)	Acquisition of portfolio investment assets (+) Repayment of portfolio investment liabilities (+)	NAFA (PI) – NIFL (PI)
	Disposal of other investment assets (-) Incurrences of other investment liabilities (-)	Acquisition of other investment assets (+) Repayment of other investment liabilities (+)	NAFA (OI) – NIFL (OI)
	Disposal of reserve assets (-) Incurrences of reserve liabilities (-)	Acquisition of reserve assets (+) Repayment of reserve liabilities (+)	NAFA (RA) – NIFL (RA) FA

BoP Accounting Identity

Given the double-entry accounting principle and the definitions of the three sub-account balances, the following extbf extcoloredBoP accounting identity holds:

$$CA + KA = FA. \quad (5)$$

Since the financial account balance measures the net lending/borrowing position of the domestic economy relative to the rest of the world, Equation (5) implies that the same must be true for the sum of the current and capital account balances:

If $CA + KA > 0$, the domestic economy is a extbf extcolorednet lender to the rest of the world.

If $CA + KA < 0$, the domestic economy is a extbf extcolorednet borrower from the rest of the world.

BoP Example: Car Export Transaction

BoP Recording of Single Market Transactions – First Example

Question:

- A German car manufacturer sells newly produced cars to a consortium of car dealers in the U.S.
- Payment is transferred from the U.S. checking account of the dealer consortium to the German manufacturer's account.
- Value of the transaction: 750,000 Euro.
- How to record this transaction in the German BoP?
- What are the balances of the German current, capital, and financial accounts?

Answer:

- Sale leads to payment from U.S. residents to German residents $\Rightarrow$ first entry is a **credit** in German BoP.
- Second entry must be a matching **debit**.
- First recording: **credit, current account, exports of goods and services**, 750,000 Euro.
- Second recording: **debit, financial account, other investment**, 750,000 Euro (increase in foreign financial asset holdings).
- BoP sub-account balances:
 - ▶ $CA = 750,000$ Euro
 - ▶ $KA = 0$
 - ▶ $FA = 750,000$ Euro
- BoP accounting identity: $CA + KA = 750,000 \text{ Euro} = FA$.

BoP Example: Stock Purchase Transaction

BoP Recording of Single Market Transactions – Second Example

Question:

- A German resident buys stock of various companies on a U.S. stock exchange.
- Payment is made in cash.
- Value of the transaction: 10,000 Euro.
- How to record this transaction in the German BoP?
- What are the balances of the German current, capital, and financial accounts?

Answer:

- Stock purchase leads to a payment from German residents to U.S. residents $\Rightarrow$ first entry is a **debit** in German BoP.
- Second entry must be a matching **credit**.
- First recording: **debit, financial account, portfolio investment**, 10,000 Euro (increase in foreign financial asset holdings).
- Second recording: **credit, financial account, other investment**, 10,000 Euro (decrease in foreign financial asset holdings).
- BoP sub-account balances:
 - ▶ $CA = 0$
 - ▶ $KA = 0$
 - ▶ $FA = 0$ (increase in stock holdings is offset by decrease in currency holdings)
- BoP accounting identity: $CA + KA = 0 = FA$.

BoP Example: Patent Purchase Funded by Foreign Loan

BoP Recording of Single Market Transactions – Third Example

Question:

- A German manufacturer of GPS devices buys a patent from a Chinese company.
- The purchase is funded by a loan from a Dutch bank.
- Value of the transaction: 200,000 Euro.
- How to record this transaction in the German BoP?
- What are the balances of the German current, capital, and financial accounts?

Answer:

- Patent purchase leads to payment from German to Chinese residents (via Dutch bank) $\Rightarrow$ first entry is a **debit** in German BoP.
- Second entry must be a matching **credit**.
- First recording: **debit, capital account, nonproduced nonfinancial assets**, –200,000 Euro (increase in foreign nonproduced asset holdings).
- Second recording: **credit, financial account, other investment**, –200,000 Euro (increase in foreign financial liability holdings).
- BoP sub-account balances:
 - ▶ $CA = 0$
 - ▶ $KA = -200,000$ Euro
 - ▶ $FA = -200,000$ Euro
- BoP accounting identity: $CA + KA = -200,000 \text{ Euro} = FA$.

Component	Germany	Euro Area	United States	China
Current account (excl. reserves)				
Goods, credit (exports)	1,728,100	3,120,700	2,020,400	3,330,900
Goods, debit (imports)	1,456,700	2,792,800	3,089,200	2,865,400
Balance on goods	271,400	327,900	-1,068,800	465,500
Services, credit (exports)	400,600	1,236,000	1,148,300	398,500
Services, debit (imports)	424,900	1,141,000	934,400	660,100
Balance on goods and services	247,100	422,900	-854,900	203,900
Primary income, credit	274,300	982,600	1,369,000	247,700
Primary income, debit	186,500	900,900	1,276,800	259,200
Balance on Primary income	334,900	504,600	-762,700	192,400
Secondary income, credit	90,100	146,700	206,400	27,900
Secondary income, debit	146,800	316,000	265,600	21,300
Current account balance (CA)	278,200	335,300	-821,900	199,000
Capital account				
Capital account, credit	59,200	51,200	170	510
Capital account, debit	58,700	77,300	7,080	710
Capital account balance (KA)	500	-26,100	-6,910	-200
Balance on current and capital account (CA + KA)	278,700	309,200	-828,800	198,800

Source: IMF (2024), data.imf.org. All figures in millions of USD.

Vietnam, BOP Analytic Presentation (Millions of U.S. Dollars)

	2021	2022	2023	Q1 2024	Q2 2024	Q3 2024
Current account (excludes reserves and related items)	-4,628.0	1,402.0	25,833.0	5,506.0	4,510.0	7,778.0
Goods, credit (exports)	335,916.0	371,715.0	355,231.0	92,881.0	97,854.0	108,235.0
Goods, debit (imports)	318,450.0	342,317.0	310,787.0	80,996.0	89,305.0	94,615.0
Balance on goods	17,466.0	29,398.0	44,444.0	11,885.0	8,549.0	13,620.0
Services, credit (exports)	4,210.0	13,526.0	19,869.0	5,758.0	5,491.0	5,873.0
Services, debit (imports)	19,877.0	27,429.0	29,072.0	7,798.0	8,308.0	9,840.0
Balance on goods and services	1,799.0	15,495.0	35,241.0	9,845.0	5,732.0	9,653.0
Primary income, credit	982.0	2,321.0	4,551.0	1,385.0	1,404.0	1,355.0
Primary income, debit	17,731.0	22,049.0	27,014.0	9,035.0	5,355.0	6,106.0
Balance on goods, services, and primary income	-14,950.0	-4,233.0	12,778.0	2,195.0	1,781.0	4,902.0
Secondary income, credit	13,568.0	12,287.0	16,112.0	4,318.0	3,800.0	3,876.0
Secondary income, debit	3,246.0	6,652.0	3,057.0	1,007.0	1,071.0	1,000.0
Capital account (excludes reserves and related items)						
Capital account, credit						
Capital account, debit						
Balance on current and capital account	-4,628.0	1,402.0	25,833.0	5,506.0	4,510.0	7,778.0

	Vietnam, BOP Analytic Presentation (Millions of U.S. Dollars)				
	2021	2021	2021	2021	2021
Financial account (excludes reserves and related items)	-30,856.3	-9,470.0	2,842.0	-1,492.0	6,262.0
<i>Direct investment, assets</i>	319.0	2,674.0	-1,550.0	80.0	120.0
Equity and investment fund shares					
Debt instruments					
<i>Direct investment, liabilities</i>	15,660.0	17,900.0	18,500.0	3,630.0	4,970.0
Equity and investment fund shares	13,868.0	15,020.0	17,753.0	3,205.0	4,341.0
Debt instruments	1,792.0	2,880.0	747.0	425.0	629.0
<i>Portfolio investment, assets</i>	-10.0	-2.0	34.0	0.0	-38.0
Equity and investment fund shares					
Debt instruments					
<i>Portfolio investment, liabilities</i>	271.0	1,510.0	-1,155.0	-529.0	-1,614.0
Equity and investment fund shares					
Debt instruments					
<i>Financial derivatives (other than reserves) and employee stock options</i>					
Fin. derivatives and employee stock options, assets					
Fin. derivatives and employee stock options, liabilities					
<i>Other investment, assets</i>	627.0	13,663.0	14,341.0	-247.0	5,150.0
Other equity					
Debt instruments	627.0	13,663.0	14,341.0	-247.0	5,150.0
<i>Other investment, liabilities</i>	15,861.3	6,395.0	-7,362.0	-1,776.0	-4,386.0
Other equity					
Debt instruments	15,861.3	6,395.0	-7,362.0	-1,776.0	-4,386.0
Balance on current, capital, and financial account	26,228.3	10,872.0	22,991.0	6,998.0	-1,752.0
Net errors and omissions	-10,365.5	-33,610.8	-17,361.4	-8,369.2	-4,314.2
Reserves and related items	15,862.9	-22,738.8	5,629.6	-1,371.2	-6,066.2
Reserve assets	15,862.9	-22,738.8	5,629.6	-1,371.2	-6,066.2
Net credit and loans from the IMF (excluding reserve position)	0.0	0.0	0.0	0.0	0.0
Exceptional financing					

As for individual households and firms, the economic well-being of an economy beyond income also depends on its wealth. A firm, for example, is routinely assessed both on the basis of its income statement and its balance sheet (the latter listing assets and liabilities). This is as measuring changes in wealth permits monitoring the sustainability of economic development.

How to measure the **national wealth** of an economy, and how does national saving contribute to the accumulation of national wealth?

Private Saving

Saving of the private sector:

$$S^{pvt} = \underbrace{Y + BPI}_{\text{GNI}} + BSI - T + GT - C \quad (6)$$

$$\underbrace{\hspace{10em}}_{\text{GNDI}}$$

$$\underbrace{\hspace{15em}}_{\text{Private GNDI}}$$

where

- T = taxes levied by government on private sector
- GT = net transfers from government to private sector

Government Saving

Saving of the government sector:

$$S^{govt} = \underbrace{T - GT}_{\text{net government income}} - G \quad (7)$$

Adding up private saving and government saving, we obtain

$$\begin{aligned} S &= S^{pvt} + S^{govt} \\ &= (Y + BPI + BSI - T + GT - C) + (T - GT - G) \\ &= Y + BPI + BSI - C - G \end{aligned} \tag{8}$$

Substituting the expenditure approach-based definition of GDP from Equation (1),

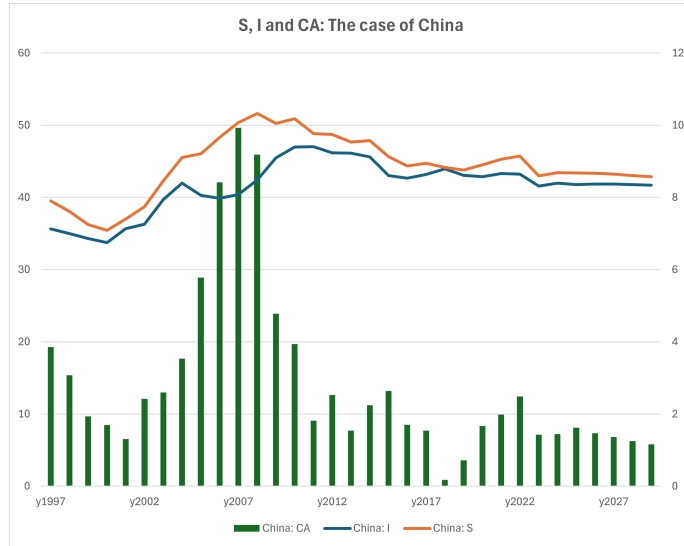
$$Y = C + I + G + TB,$$

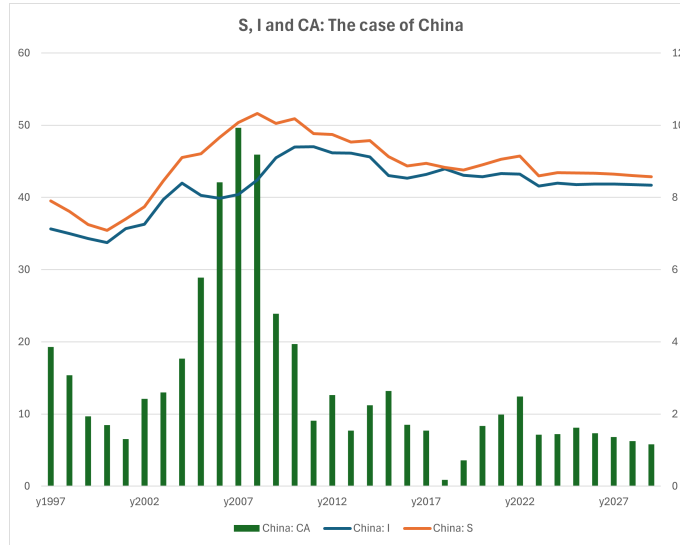
into Equation (8), we obtain:

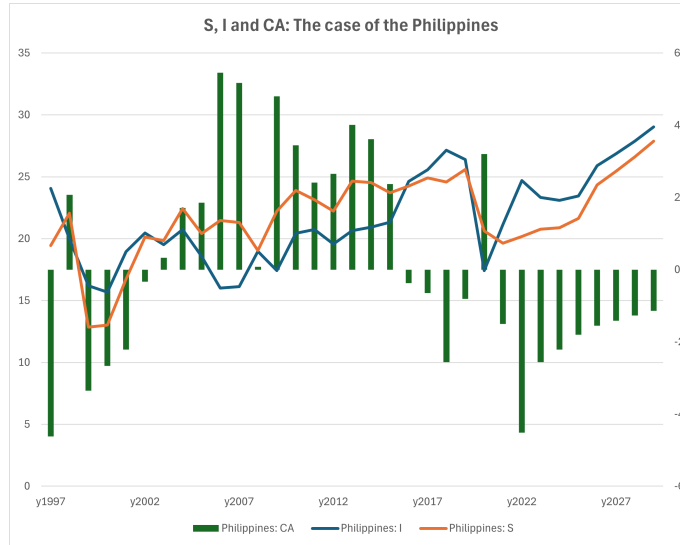
$$S = I + \underbrace{TB + BPI + BSI}_{CA} \tag{9}$$

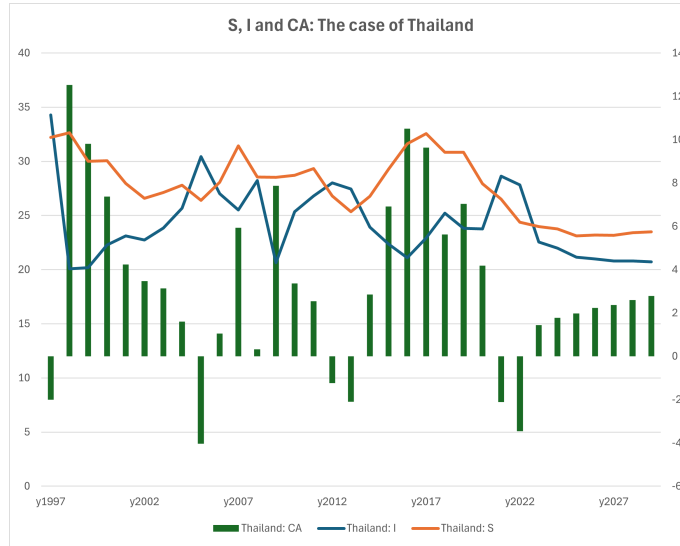
National saving can thus be allocated to two uses:

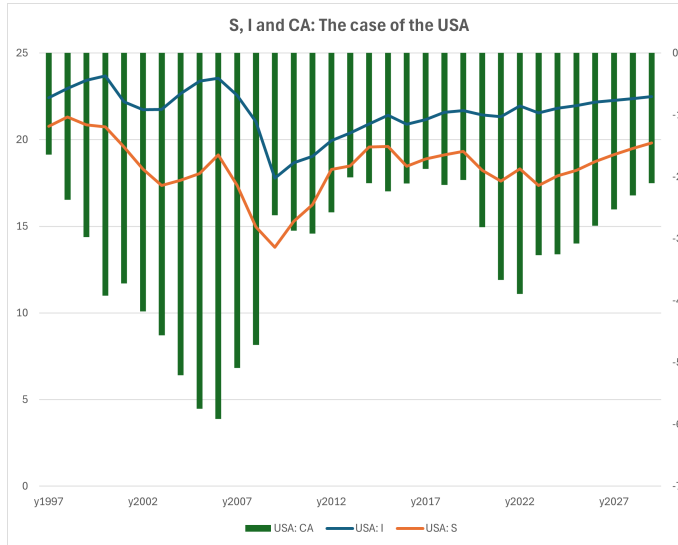
- (i) funding (through I) the domestic accumulation of physical capital;
- (ii) funding the current account balance, that is (see our discussion of the BoP), being a net lender to the rest of the world (if $CA > 0$, and under $KA = 0$).

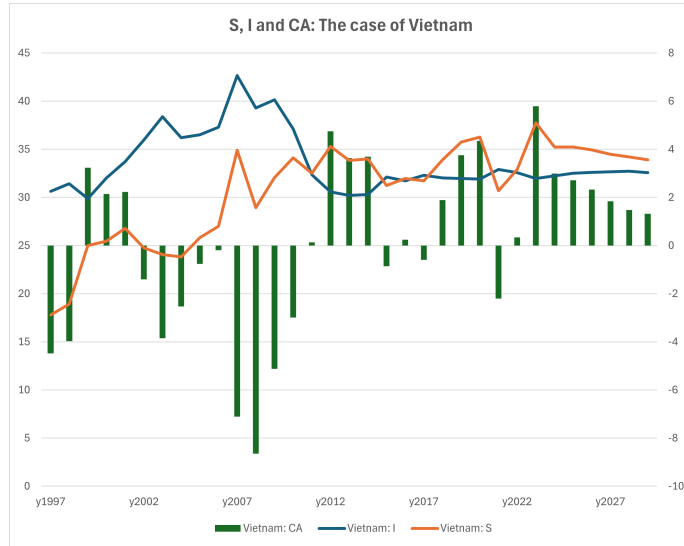












National wealth is the overall wealth of the residents of an economy.

To measure it, consider first the net asset position *of an individual*. At the end of period t :

$$\begin{aligned}\text{Net asset position}_t &= \text{Net asset position}_{t-1} \\ &\quad + \text{Acquisitions of assets}_t \\ &\quad - \text{Disposals of assets}_t \\ &\quad + \text{Valuation changes}_t\end{aligned}$$

In general, end-of-period wealth is a function of:

- beginning-of-period wealth,
- saving (net asset acquisitions minus net asset disposals),
- valuation changes.

Tracing saving sufficiently far back into the past, the components of saving (including their subsequent valuation changes) thus inform us about the components of wealth.

From Equation (9), we know that national saving can be allocated either towards investing in physical capital in the domestic economy or towards lending to the rest of the world, that is, towards accumulation of wealth abroad.

Note that the domestic economy's stock of physical capital (K) is related to investment in physical capital as follows:

$$K_{\text{end of period } t} = K_{\text{beginning of period } t} - \underbrace{\delta}_{\substack{\text{rate of} \\ \text{depreciation} \\ \text{of physical}}} \cdot K_{\text{beginning of period } t}$$

National wealth is composed of:

- Domestic capital stock (K),
- Wealth abroad: **net international investment position (NIIP)**, which equals foreign assets (FOA) minus foreign liabilities (FOL).

$$\text{national wealth} = \underbrace{K}_{\text{domestic wealth}} + \underbrace{FOA - FOL}_{\text{wealth abroad (NIIP)}} \quad (11)$$

Notes:

- Domestic financial assets held by domestic residents are *not* part of national wealth.
- The value of these assets is offset by equivalent domestic liabilities.
- Equating the stock of physical capital at the end of period t with the stock at the beginning of period $t + 1$, the capital accumulation equation, (10), may also be written as:

$$\begin{aligned} K_{t+1} &= K_t - \delta \cdot K_t + I_t \\ &= (1 - \delta) \cdot K_t + I_t \end{aligned} \quad (12)$$

Equation (12) will feature in our short-, medium- and long-run models of the macroeconomy. Let us further illustrate national wealth accounting by considering the calculation of national saving and national wealth for two sample economies.

Calculating National Saving and National Wealth for Sample Economies

Green Engine GmbH is located in the economy East, and at the beginning of the year 2020 had a capital stock worth 70,000 Euro. During 2020, Green Engine builds 200 electric engines and its capital stock depreciates by 4,000 Euro. The market value of each engine is 300 Euro. Green Engine's costs for the production of each engine are: cost of labor from workers who are residents of East, 20 Euro; cost of labor from workers who are residents of the economy West, 75 Euro; taxes, 10 Euro. Green Engine sells all 200 of the engines produced in 2020 to Quiet Vehicles AG (which is located in West). Also during 2020, the owners of Green Engine purchase 3,000 shares of the stock of Quiet Vehicles AG, at a price of 10 Euro per share. Finally during 2020, the owners of Green Engine earn 5,000 Euro in interest payments on a savings account held with a bank in West (the balance on this account at the beginning of the year 2020 had been 75,000 Euro, and the only transaction in this account that occurs during 2020 is the crediting of the 5,000 Euro interest payment).

Quiet Vehicles entered the year 2020 with a capital stock worth 250,000 Euro. During 2020, Quiet Vehicles builds 200 motorcycles and its capital stock depreciates by 20,000 Euro. The costs for the production of each motorcycle are: 300 Euro worth of components from Green Engine; 500 Euro cost of labor from workers who are residents of West. Quiet Vehicles sells 5 of the motorcycles produced for 2,000 Euro to private households in East; it sells 105 of the motorcycles produced to private households in West (also at 2,000 Euro per motorcycle). At the end of the year 2020, Quiet Vehicles puts all remaining 90 motorcycles produced but not sold in 2020 into its inventory.

How to calculate the year 2020 national saving as well as the national wealth at the end of the year 2020 of the economies East and West?

National Saving:

	East	West
Investment	0	90 · 800 (addition to inventory)
Trade Balance	200 · 300 (exports of engines) – 5 · 2,000 (imports of motorcycles)	– 200 · 300 (imports of engines) + 5 · 2,000 (exports of motorcycles)
Balance on Primary Income	– 200 · 75 (wages paid to workers from West) + 5,000 (interest income earned in West)	+ 200 · 75 (wages of workers earned in East) – 5,000 (interest income paid to East)
Balance on Secondary Income	0	0
Total = National Saving	40,000	32,000

National Wealth:

	East	West
Capital Stock	70,000 (initial capital stock) – 4,000 (depreciation)	250,000 (initial capital stock) – 20,000 (depreciation) + 90 · 800 (addition to capital stock)
Foreign Nonfinancial and Financial Assets	3,000 · 10 (stocks held in West) 75,000 + 5,000 (savings account held in West)	
Foreign Nonfinancial and Financial Liabilities	0	3,000 · 10 (stocks held by East) 75,000 + 5,000 (savings account held by East)
Total = National Wealth	176,000	192,000

Potential Problems with this Measure of National Wealth

As a measure of an economy's comprehensive wealth, the wealth measure we have considered appears to omit relevant factors, including

- **“human capital”** (the education and skills of the economy's people), and
- **“natural capital”** (including the economy's land, forests, minerals and energy).

Alternative measures of an economy's wealth do exist, for example the World Bank's **“total wealth”** that combines physical, human and natural capital as well as net foreign assets, but data coverage is still relatively limited.

Country	Wealth Indicators	y2019	y2020
United States	National comprehensive wealth (current US\$)	247,143,761,900,769	246,222,814,588,133
China	National comprehensive wealth (current US\$)	113,657,626,599,143	118,987,819,854,739
Germany	National comprehensive wealth (current US\$)	48,784,054,584,478	50,256,065,079,481
Thailand	National comprehensive wealth (current US\$)	4,788,250,324,861	4,536,014,015,367
Viet Nam	National comprehensive wealth (current US\$)	3,117,219,892,698	3,195,880,318,898
Philippines	National comprehensive wealth (current US\$)	2,825,890,115,168	2,770,211,518,186
United States	National comprehensive wealth per capita (current US\$)	752,730	742,693
Germany	National comprehensive wealth per capita (current US\$)	587,102	604,323
China	National comprehensive wealth per capita (current US\$)	80,737	84,323
Thailand	National comprehensive wealth per capita (current US\$)	67,149	63,462
Viet Nam	National comprehensive wealth per capita (current US\$)	32,547	33,067
Philippines	National comprehensive wealth per capita (current US\$)	25,601	24,692
China	Foreign assets per capita (current US\$)	5,501.18	6,194.30
Germany	Foreign assets per capita (current US\$)	128,278.60	152,199.28
Philippines	Foreign assets per capita (current US\$)	1,726.39	2,009.66
Thailand	Foreign assets per capita (current US\$)	7,529.70	8,362.75
United States	Foreign assets per capita (current US\$)	86,721.87	95,160.19
Viet Nam	Foreign assets per capita (current US\$)	1,682.69	1,929.40
China	Foreign liabilities per capita (current US\$)	3,960.02	4,662.61
Germany	Foreign liabilities per capita (current US\$)	102,766.60	122,435.71
Philippines	Foreign liabilities per capita (current US\$)	2,123.39	2,265.73
Thailand	Foreign liabilities per capita (current US\$)	7,971.14	7,968.01
United States	Foreign liabilities per capita (current US\$)	123,419.77	141,011.31
Viet Nam	Foreign liabilities per capita (current US\$)	3,171.28	3,349.10

4. Real vs. Nominal GDP

- Before discussing these measures, just have a some brief look at these statistics:
 - In the year 1985, a lunch portion similar to that at the VGU canteen cost about 4 VND. Now you pay about 20,000 VND.
 - In the same year 1985, the lunch would cost 0.4 USD. Now you pay roughly 0.78 USD.

$$(\text{Nominal GDP})_t = \sum_t \underbrace{p_{it}}_{\substack{\text{price of} \\ \text{good/ser-} \\ \text{vice } i \text{ in} \\ \text{year } t}} \cdot \underbrace{q_{it}}_{\substack{\text{quan-} \\ \text{tity of} \\ \text{good/ser-} \\ \text{vice } i \text{ in} \\ \text{year } t}},$$

When comparing values of nominal GDP across different years, this generally involves both changes in quantities and prices, and thus does not allow to infer about changes in the standard of living. To allow for such inference, real GDP as traditionally measured adjusts for price changes and reflects the sum of the value added produced by all firms in the country in terms of only one set of prices, namely those of the so-called base year.

Real vs. Nominal GDP

- The **Laspeyres** approach to computing real GDP involves using initial year prices as the base-year prices:

$$(\text{Real GDP})_t^{\text{Laspeyres}} = Y_t^{\text{Laspeyres}} = \sum_{i=1}^N p_{i0} \times q_{it}, t = 0, 1, \dots, T,$$

implying the growth rate of real GDP:

$$1 + g_{Y_t}^{\text{Laspeyres}} = \frac{\sum_{i=1}^N p_{i0} \times q_{it}}{\sum_{i=1}^N p_{i0} \times q_{i,t-1}}, t = 0, 1, \dots, T,$$

The main problem with the Laspeyres approach is that as one moves from the initial year ($t = 0$) to the final year ($t = T$), Laspeyres-based real GDP over-states more and more the contribution to GDP of goods and services that are becoming less and less expensive.

Real vs. Nominal GDP

- The **Paasche** approach to computing real GDP involves using final year prices as the base-year prices:

$$(\text{Real GDP})_t^{\text{Laspeyres}} = Y_t^{\text{Laspeyres}} = \sum_{i=1}^N p_{iT} \times q_{it}, t = 0, 1, \dots, T,$$

implying the growth rate of real GDP:

$$1 + g_{Y_t}^{\text{Laspeyres}} = \frac{\sum_{i=1}^N p_{iT} \times q_{it}}{\sum_{i=1}^N p_{iT} \times q_{i,t-1}}, t = 0, 1, \dots, T,$$

The main problem with the Paasche approach is that as, starting in the final year ($t = T$), one moves back in time towards the initial year ($t = 0$), Paasche-based real GDP over-states more and more the contribution to GDP of goods and services for which the price is continuously rising from the initial year to the final year.

Real vs. Nominal GDP

The **Annual Chain Weight** approach aims to overcome these (opposing) biases of the Laspeyres and Paasche approaches and computes real GDP using the geometric average of results for the Laspeyres and Paasche approaches, with annually updated base years:

$$1 + g_{Y_t}^{\text{Annual Chain Weight}} = \sqrt{\underbrace{\frac{\sum_{i=1}^N p_{i,t-1} \times q_{it}}{\sum_{i=1}^N p_{i,t-1} \times q_{i,t-1}}}_{\text{Annual Laspeyres}} \times \underbrace{\frac{\sum_{i=1}^N p_{i,t} \times q_{it}}{\sum_{i=1}^N p_{i,t} \times q_{i,t-1}}}_{\text{Annual Paasche}}},$$

$$t = 0, 1, \dots, T.$$

To start with,

$$(\text{Real GDP})_0^{\text{Annual Chain Weight}} = Y_0^{\text{Annual Chain Weight}} = \sum_{i=1}^N p_{i0} \times q_{i0}$$

and,

$$Y_t^{\text{Annual Chain Weight}} = (1 + g_{Y_t}^{\text{Annual Chain Weight}}) \times Y_{t-1}^{\text{Annual Chain Weight}}$$

In the country Sol, the following number of (newly produced) chicken wraps and smart phones were sold in 2023, 2024 and 2025 (no other goods or services were produced or sold):

	2023 Quantity	2023 Price (Euro)	2024 Quantity	2024 Price (Euro)	2025 Quantity	2025 Price (Euro)
Chicken Wraps	4000	2.5	4000	2.55	4000	2.6
Smart Phones	3	650	6	600	9	550

Calculate nominal and real GDP (the latter using the Laspeyres, Paasche and Annual Chain Weight approaches) in Sol for 2023, 2024 and 2025. What explains the differences, if any, between the real GDP numbers under the three different approaches?

Nominal and Real GDP Calculations

We then readily obtain ...

... using Equation (12) for nominal GDP:

$$(\text{Nominal GDP})_{2023} = 2.50 \times 4,000 + 650 \times 3 = 11,950$$

$$(\text{Nominal GDP})_{2024} = 2.55 \times 4,000 + 600 \times 6 = 13,800$$

$$(\text{Nominal GDP})_{2025} = 2.60 \times 4,000 + 550 \times 9 = 15,350$$

... using Equation (13) for Laspeyres-based real GDP:

$$Y_{2023}^{\text{Laspeyres}} = 2.50 \times 4,000 + 650 \times 3 = 11,950$$

$$Y_{2024}^{\text{Laspeyres}} = 2.50 \times 4,000 + 650 \times 6 = 13,900$$

$$Y_{2025}^{\text{Laspeyres}} = 2.50 \times 4,000 + 650 \times 9 = 15,850$$

Nominal and Real GDP Calculations

We then readily obtain ...

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... using Equation (13) for Laspeyres-based real GDP:

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$$Y_{2025}^{\text{Laspeyres}} = 2.50 \times 4,000 + 650 \times 9 = 15,850$$

Real GDP: Paasche and Chain Weight Methods

... using Equation (15) for Paasche-based real GDP:

$$Y_{2023}^{\text{Paasche}} = 2.60 \times 4,000 + 550 \times 3 = 12,050$$

$$Y_{2024}^{\text{Paasche}} = 2.60 \times 4,000 + 550 \times 6 = 13,700$$

$$Y_{2025}^{\text{Paasche}} = 2.60 \times 4,000 + 550 \times 9 = 15,350$$

... using Equations (17), (18) and (19) for Annual Chain Weight-based real GDP:

$$Y_{2023}^{\text{Annual Chain Weight}} = 2.50 \times 4,000 + 650 \times 3 = 11,950$$

$$Y_{2024}^{\text{Annual Chain Weight}} = \sqrt{\frac{2.50 \times 4,000 + 650 \times 6}{2.50 \times 4,000 + 650 \times 3} \times \frac{2.55 \times 4,000 + 600 \times 6}{2.55 \times 4,000 + 600 \times 3}} \times Y_{2023}^{\text{Annual Chain Weight}}$$

$$Y_{2025}^{\text{Annual Chain Weight}} = \sqrt{\frac{2.55 \times 4,000 + 600 \times 9}{2.55 \times 4,000 + 600 \times 6} \times \frac{2.60 \times 4,000 + 550 \times 9}{2.60 \times 4,000 + 550 \times 6}} \times Y_{2024}^{\text{Annual Chain Weight}}$$

Real GDP: Paasche and Chain Weight Methods

... using Equation (15) for Paasche-based real GDP:

$$Y_{2023}^{\text{Paasche}} = 2.60 \times 4,000 + 550 \times 3 = 12,050$$

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... using Equations (17), (18) and (19) for Annual Chain Weight-based real GDP:

$$Y_{2023}^{\text{Annual Chain Weight}} = 2.50 \times 4,000 + 650 \times 3 = 11,950$$

$$Y_{2024}^{\text{Annual Chain Weight}} = \sqrt{\frac{2.50 \times 4,000 + 650 \times 6}{2.50 \times 4,000 + 650 \times 3} \times \frac{2.55 \times 4,000 + 600 \times 6}{2.55 \times 4,000 + 600 \times 3}} \times Y_{2023}^{\text{Annual Chain Weight}}$$

$$Y_{2025}^{\text{Annual Chain Weight}} = \sqrt{\frac{2.55 \times 4,000 + 600 \times 9}{2.55 \times 4,000 + 600 \times 6} \times \frac{2.60 \times 4,000 + 550 \times 9}{2.60 \times 4,000 + 550 \times 6}} \times Y_{2024}^{\text{Annual Chain Weight}}$$

Explaining Differences in Real GDP Measures

The Laspeyres and Annual Chain Weight approaches to calculating real GDP involve the same initializations, and thus the same values for real GDP in 2023 (which in turn are equal to nominal GDP in 2023). The Paasche approach, in contrast, sets nominal GDP equal to real GDP for the final year, here 2025. This alone drives some of the differences in the real GDP numbers.

The Laspeyres approach-based real GDP both for 2024 and 2025 is higher than the Annual Chain Weight approach-based real GDP, as for both years the smart phone transactions are over-valued under the Laspeyres approach. The over-valuation is even stronger for 2025 than for 2024, and for this reason $\gamma_{2025}^{\text{Laspeyres}}$ exceeds $\gamma_{2025}^{\text{Annual Chain Weight}}$ by 1.90%, whereas the difference between $\gamma_{2024}^{\text{Laspeyres}}$ and $\gamma_{2024}^{\text{Annual Chain Weight}}$ is “only” 0.57%.

In comparison to the Annual Chain Weight approach-based real GDP, the Paasche approach-based real GDP both for 2023 and 2024 under-values the smart phone transactions, leading to smaller values of real GDP.

Having computed nominal and real GDP levels, we can also obtain the (so-called GDP deflator-based) **price level** as:

$$(\text{Price Level})_t = P_t = \frac{(\text{Nominal GDP})_t}{(\text{Real GDP})_t},$$

and the (GDP deflator-based) **rate of inflation** as:

$$(\text{Rate of Inflation})_t = \pi_t \approx (\text{Percentage change in nominal GDP})_{\text{from } t-1 \text{ to } t} \\ - (\text{Percentage change in real GDP})_{\text{from } t-1 \text{ to } t}$$

Inflation Calculations by Approach

... for the Laspeyres approach:

$$\pi_{2024}^{\text{Laspeyres}} = \left(\frac{13,800 - 11,950}{11,950} - \frac{13,900 - 11,950}{11,950} \right) \times 100\% = -0.84\%$$

$$\pi_{2025}^{\text{Laspeyres}} = \left(\frac{15,350 - 13,800}{13,800} - \frac{15,850 - 13,900}{13,900} \right) \times 100\% = -2.80\%$$

... for the Annual Chain Weight approach:

$$\pi_{2024}^{\text{Annual Chain Weight}} = \left(\frac{13,800 - 11,950}{11,950} - \frac{13,821.03 - 11,950}{11,950} \right) \times 100\% = -0.18\%$$

$$\pi_{2025}^{\text{Annual Chain Weight}} = \left(\frac{15,350 - 13,800}{13,800} - \frac{15,554.53 - 13,821.03}{13,821.03} \right) \times 100\% = -1.31\%$$

GDP Deflator-Based Price Level and Inflation

Having computed nominal and real GDP levels, we can also obtain the (so-called GDP deflator-based) **price level** as:

$$(\text{Price Level})_t = P_t = \frac{(\text{Nominal GDP})_t}{(\text{Real GDP})_t},$$

and the (GDP deflator-based) **rate of inflation** as:

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Inflation Calculations by Approach

... for the Laspeyres approach:

$$\pi_{2024}^{\text{Laspeyres}} = \left(\frac{13,800 - 11,950}{11,950} - \frac{13,900 - 11,950}{11,950} \right) \times 100\% = -0.84\%$$

$$\pi_{2025}^{\text{Laspeyres}} = \left(\frac{15,350 - 13,800}{13,800} - \frac{15,850 - 13,900}{13,900} \right) \times 100\% = -2.80\%$$

... for the Annual Chain Weight approach:

$$\pi_{2024}^{\text{Annual Chain Weight}} = \left(\frac{13,800 - 11,950}{11,950} - \frac{13,821.03 - 11,950}{11,950} \right) \times 100\% = -0.18\%$$

$$\pi_{2025}^{\text{Annual Chain Weight}} = \left(\frac{15,350 - 13,800}{13,800} - \frac{15,554.53 - 13,821.03}{13,821.03} \right) \times 100\% = -1.31\%$$



T7/2021 so với tháng T6/2021



T7/2021 so với tháng T7/2020

- Consumer price index (CPI) measures the overall level of the prices of the goods and services consumed by a typical household. It is a measure of the overall cost of goods and services bought by a typical consumer. Computed and reported every month by the General Statistics Office.
- GDP Deflator: A measure of the overall price of all goods and services in GDP, including investment goods, public goods (public security, national defense) and export goods and services.



- 1. Fix the basket: Which prices are most important to the typical consumer? Different weight
- 2. Find the prices: At each point in time
- 3. Compute the basket's cost: Same basket of goods $\Rightarrow$ Isolate the effects of price changes
- 4. Choose a base year and compute the CPI: Base year = benchmark. Price of a basket of goods and services in current year divided by price of the basket in base year times 100
- 5. Compute the inflation rate: $Inflation\ in\ year\ t = \frac{old/CPI_t - old/CPI_{t-1}}{old/CPI_{t-1}} \times 100\%$

Step 1: Fixed Basket of Goods

Basket = 4 hot dogs, 2 hamburgers

Step 2: Price of Each Good

Year	Hot Dogs	Hamburgers
2019	\$1	\$2
2020	2	3
2021	3	4

Step 3: Cost of Basket

- 2019: $(1 \times 4) + (2 \times 2) = \8
- 2020: $(2 \times 4) + (3 \times 2) = \14
- 2021: $(3 \times 4) + (4 \times 2) = \20

Step 4: CPI (Base = 2019)

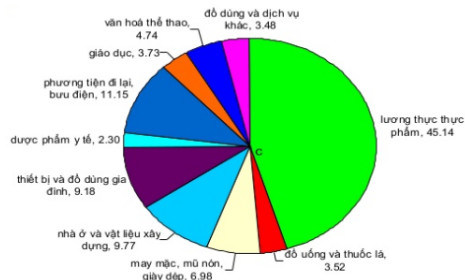
- 2019: $(8/8) \times 100 = 100$
- 2020: $(14/8) \times 100 = 175$
- 2021: $(20/8) \times 100 = 250$

Step 5: Inflation Rate

- 2020: $\frac{175 - 100}{100} \times 100 = 75\%$
- 2021: $\frac{250 - 175}{175} \times 100 = 43\%$

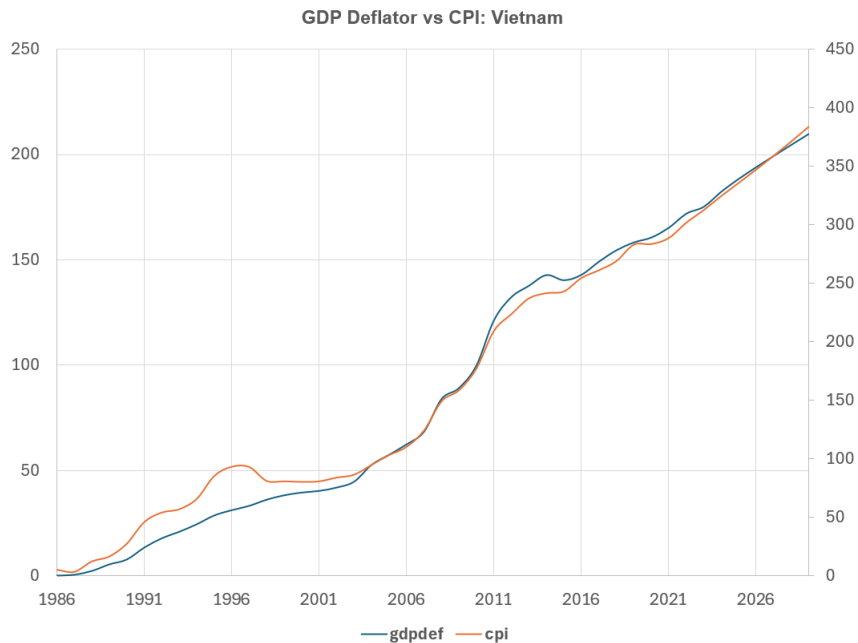
32. Chỉ số giá tiêu dùng, chỉ số giá vàng, chỉ số giá đô la Mỹ và lạm phát cơ bản tháng 6 năm 2022

	Tháng 6 năm 2022 so với					
	Kỳ gốc	Tháng 6	Tháng 12	Tháng 5	Bình quân	Bình quân
	(2019)	năm	năm	năm	quý II	6 tháng
		2021	2021	2022	năm 2022	năm 2022
					so với	so với
					cùng kỳ	cùng kỳ
					năm trước	năm trước
CHỈ SỐ GIÁ TIÊU DÙNG						
Hàng ăn và dịch vụ ăn uống	111.49	102.27	102.31	100.80	101.56	100.89
Trong đó: Lương thực	110.95	102.87	101.26	100.16	102.45	102.32
Thực phẩm	110.66	101.29	102.00	100.98	100.41	99.60
Ăn uống ngoài gia đình	113.81	104.48	103.53	100.65	104.04	103.50
Đồ uống và thuốc lá	106.64	103.22	101.90	100.35	102.98	102.76
May mặc, mũ nón và giày dép	103.81	101.49	100.91	100.17	101.32	101.09
Nhà ở và vật liệu xây dựng	106.51	101.53	102.94	99.99	102.14	102.01
Thiết bị và đồ dùng gia đình	104.27	101.99	101.40	100.31	101.80	101.55
Thuốc và dịch vụ y tế	102.72	100.37	100.21	100.04	100.35	100.31
Trong đó: Dịch vụ y tế	102.46	100.04	100.02	100.01	100.03	100.03
Giáo dục	121.42	121.41	114.42	103.62	118.79	117.43
Bưu chính viễn thông	97.66	99.50	99.68	99.84	99.55	99.45
Giáo dục	104.92	97.87	101.83	100.07	97.79	97.18
Trong đó: Dịch vụ giáo dục	104.69	97.21	101.94	100.03	97.14	96.44
Văn hoá, giải trí và du lịch	102.04	103.40	103.35	100.52	102.66	101.43
Hàng hóa và dịch vụ khác	107.17	102.21	101.68	100.24	102.04	101.75



Compute the CPI for each column in this table, using the weights provided in the chart on the right.

- CPI is, again, not a perfect measure.
 - ▶ Substitution bias: Prices do not change proportionately, and consumers substitute toward goods that have become relatively less expensive
 - ▶ Introduction of new goods: more variety of goods are not captured timely as the CPI basket is fixed. The increased set of possible choices makes each dollar more valuable. The same is true with the evolution of the economy over time: As new goods are introduced, consumers have more choices, and each dollar is worth more.
 - ▶ Unmeasured quality change: Changes in quality
- Core CPI: Measure of the overall cost of consumer goods and services excluding food and energy
- Producer price index, PPI: Measure of the cost of a basket of goods and services bought by firms. Changes in PPI are often thought to be useful in predicting changes in CPI



Application of CPI index

$$\text{Amount in today dong} = \frac{\text{Price in today's dong}}{\text{Price in Year T's dong}} \times \text{Amount in year T dong}$$

- CPI is the basis for correction of monetary amounts: Automatic correction by law or contract, known as indexation.
- COLA: Cost-of-living allowance
- Nominal interest rate: Interest rate as usually reported, without a correction for the effects of inflation
- Real interest rate: Interest rate corrected for the effects of inflation = Nominal interest rate – Inflation rate

PPP GDP

- Other topics often included in a chapter on macroeconomic measurement are national wealth, the balance of payment (BoP), and purchasing power parity exchange rate. BoP will be discussed later.
- To complete the discussion, we now focus on the Purchasing Power Parity Exchange Rate-Based GDP levels.
- How to compare levels and growth rates of GDP per capita across countries featuring different currencies? This task is complicated by the fact that prices of traded goods relative to those of non-traded goods tend to be sizably higher in "poor" countries than in "rich" countries, and that pricing of market exchange rates is more closely related to the pricing of traded than of non-traded goods.
- Purchasing power parity (PPP) exchange rates use a standardized basket of traded and non-traded goods and services across the countries in question.

PPP: An example

- To illustrate the idea, let's focus on a rather realistic example:
- As of today, the exchange rate $USD/VND = 23,800$

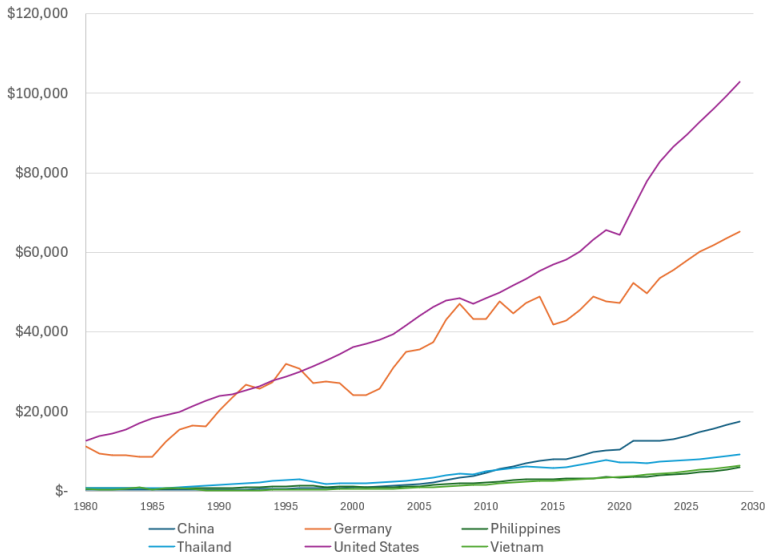
	Traded good (For example: Smart phone)		Non-traded good (For example: Student room for rent)	
	Production per Capita	Price (Local currency)	Production per Capita	Price (Local currency)
USA	20	100	300	300
Vietnam	3	2,380,000	50	1800,000

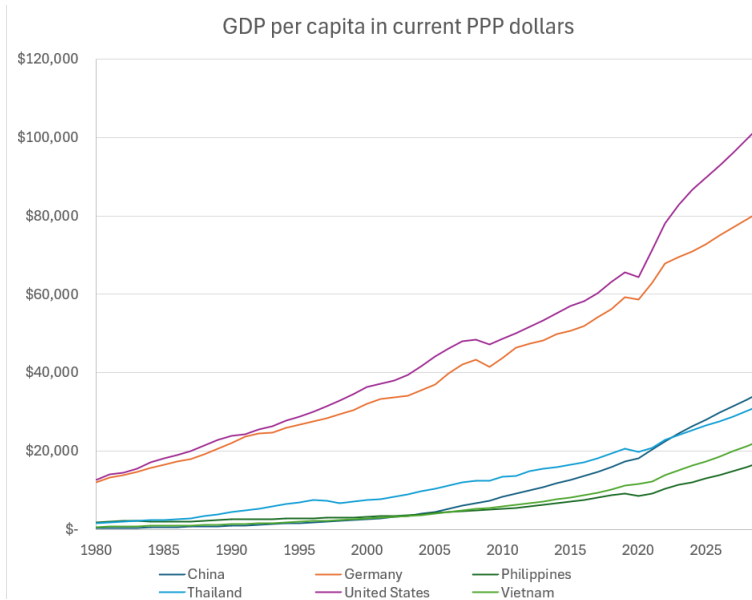
- Using the current exchange rate, GDP per capita in Vietnam is $3 \times 2,380,000 + 50 \times 1,800,000 = 97,140,000$ VND, or \$ 4,081.51, which is quite close to reality.
- The GDP per capita for the US is: $100 \times 20 + 300 \times 300 = 92,000$ USD.
- However, the market exchange rate reflects the price of tradable goods, such as smart phones. Non-tradable goods/services such as rental price might not have any influence on the market exchange rate.

PPP: An example

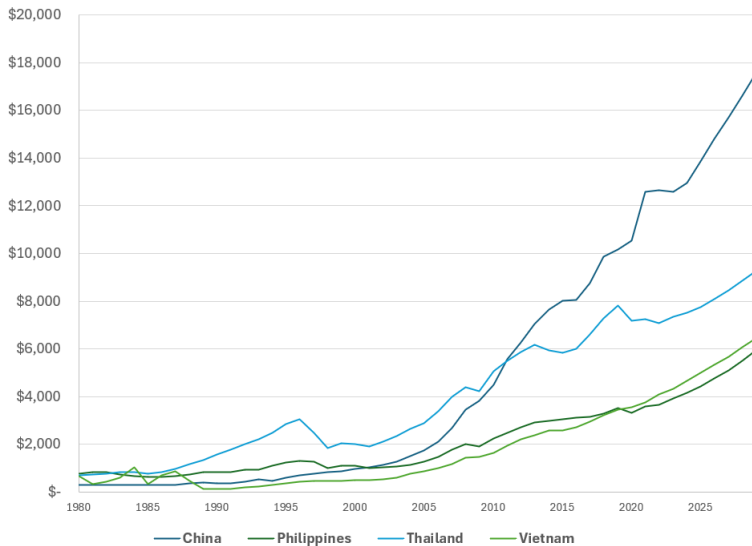
- Using market exchange rate: US people enjoy a living standard of about 22.54 times better than Vietnamese people.
- That seems questionable: computer? about 7 times. rooms? about 6 times! How can a weighted average of 6 and 7 makes 22?
- To correct for this bias, let's fix the basket at 1 smart phone and 50 rooms. The cost of this basket in Vietnam is 92,380,000 VND. The cost of this basket in the USA is 15,100 USD.
- Vietnam's PPP exchange rate is $92,380,000/15,100 = 6,117.88$, which is quite different from the current exchange rate of 23,800. Correcting for the purchasing power parity, the GDP per capita is $97,140,000/6,117.88 = 15,878.05$ PPP\$. Which is quite close to the reality.

GDP per capita in current USD

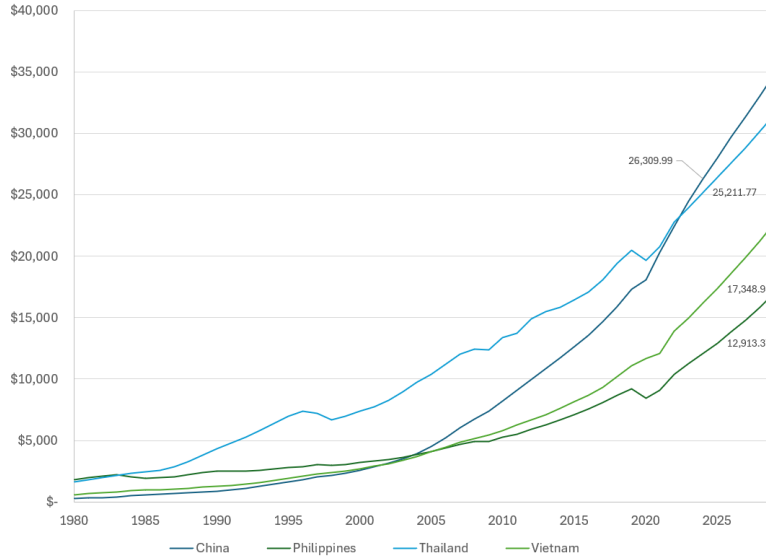




GDP per capita in current USD



GDP per capita in current PPP dollars



Make sense of international databases

- The IMF WEO
- The WB WDI
- The Fed Louis
- The IMF BOP
- The World Bank Wealth Accounts